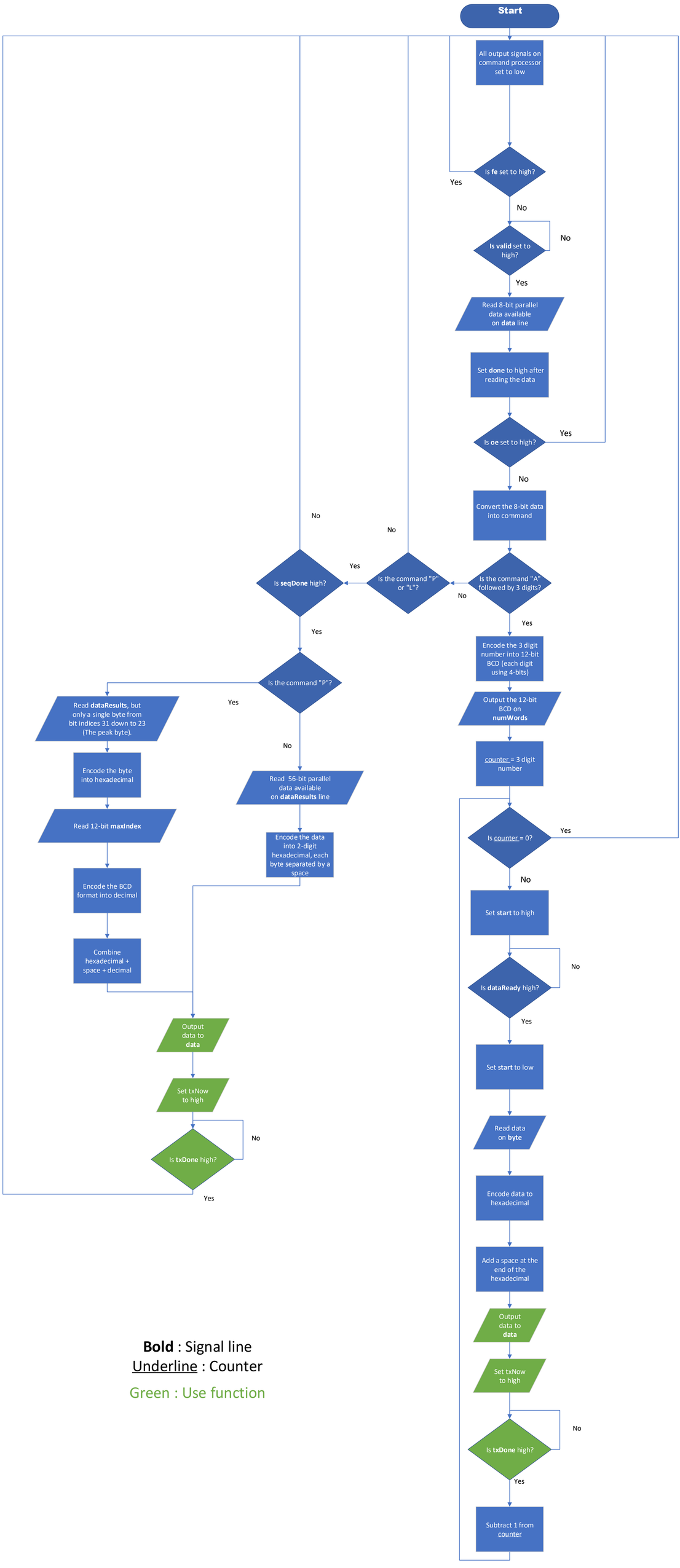

Start
All output signals on command processor set to low
Is fe set to high?
Yes
No
Is valid set to high?
No
Yes
Read 8-bit parallel data available on data line
Set done to high after reading the data
Is oe set to high?
Yes
No
Convert the 8-bit data into command
Is the command "A" followed by 3 digits?
No
Yes
Is the command "P" or "L"?
Yes
No
Is seqDone high?
No
Yes
Encode the 3 digit number into 12-bit BCD (each digit using 4-bits)
Output the 12-bit BCD on numWords
counter = 3 digit number
Is counter = 0?
Yes
No
Set start to high
Is dataReady high?
No
Yes
Set start to low
Read data on byte
Encode data to hexadecimal
Add a space at the end of the hexadecimal
Output data to data
Set txNow to high
Is txDone high?
No
Yes
Subtract 1 from counter
Is the command "P"?
Yes
No
Read dataResults, but only a single byte from bit indices 31 down to 23 (The peak byte).
Encode the byte into hexadecimal
Read 12-bit maxIndex
Encode the BCD format into decimal
Combine hexadecimal + space + decimal
Read 56-bit parallel data available on dataResults line
Encode the data into 2-digit hexadecimal, each byte separated by a space
Output data to data
Set txNow to high
Is txDone high?
No
Yes
Bold : Signal line
Underline : Counter
Green : Use function